

Paper 1B drill: linearisation

Fifteen items. What do I plot against what?

IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **25 marks**

For each relationship, state what to plot on the vertical axis and what on the horizontal axis so that the graph is a *straight line*, then say what the *gradient* represents. Where the line does not pass through the origin, say what the *intercept* represents too. Aim for the form $y = mx + c$.

A substitutions: plot a power or a reciprocal

A1. A pendulum: $T = 2\pi\sqrt{L/g}$, with L varied. [2]

A2. A gas at constant temperature: $pV = \text{constant}$, with V varied. [2]

A3. Uniform acceleration from rest: $s = \frac{1}{2}at^2$. [2]

A4. Newton's law of gravitation: $F = \frac{GMm}{r^2}$, with r varied. [2]

A5. A black body: $P = \sigma AT^4$, with T varied. [1]

B logarithms

B1. Radioactive decay: $N = N_0e^{-\lambda t}$. [2]

B2. Absorption of gamma rays: $I = I_0e^{-\mu x}$. [2]

B3. Capacitor discharge: $V = V_0e^{-t/RC}$. [2]

B4. A power law $y = Ax^n$ in which the exponent n is *unknown*. [2]

C the intercept matters too

C1. Uniform acceleration: $v^2 = u^2 + 2as$, with s varied. [2]

C2. The photoelectric equation: $E_{\text{max}} = hf - \phi$, with f varied. [2]

C3. Uniform acceleration with a head start: $s = ut + \frac{1}{2}at^2$. Divide through by t first. [2]

D getting the constant out

D1. In A1, the gradient is measured as $4.05 \text{ s}^2 \text{ m}^{-1}$. Determine g . [2]

D2. A resonance tube obeys $L = \frac{v}{4}\left(\frac{1}{f}\right) - e$, where e is the end correction. State what to plot, and what both the gradient and the intercept give. [2]

D3. A thin lens obeys $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$. State what to plot, and explain why the gradient carries no physics here while the intercept does. [2]

Answers

A *substitutions*

- A1. Squaring gives $T^2 = \frac{4\pi^2}{g}L$. Plot T^2 against L ; gradient = $\frac{4\pi^2}{g}$; through the origin.
- A2. $p = \text{constant} \times \frac{1}{V}$. Plot p against $1/V$; gradient = the constant, which is nRT ; through the origin.
- A3. Plot s against t^2 ; gradient = $\frac{1}{2}a$, so $a = 2 \times \text{gradient}$; through the origin.
- A4. Plot F against $1/r^2$; gradient = GMm ; through the origin.
- A5. Plot P against T^4 ; gradient = σA ; through the origin.
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B *logarithms*

- B1. Taking natural logs, $\ln N = \ln N_0 - \lambda t$. Plot $\ln N$ against t ; gradient = $-\lambda$; intercept = $\ln N_0$.
- B2. $\ln I = \ln I_0 - \mu x$. Plot $\ln I$ against x ; gradient = $-\mu$; intercept = $\ln I_0$.
- B3. $\ln V = \ln V_0 - \frac{1}{RC}t$. Plot $\ln V$ against t ; gradient = $-\frac{1}{RC}$, so $RC = -1/\text{gradient}$; intercept = $\ln V_0$.
- B4. $\log y = \log A + n \log x$. Plot $\log y$ against $\log x$; gradient = n , the exponent itself; intercept = $\log A$. This is the one case where logarithms are *necessary* rather than merely convenient: with n unknown you cannot choose what power to plot against, so you let the gradient tell you.
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C *the intercept matters*

- C1. Plot v^2 against s ; gradient = $2a$; intercept = u^2 , so the initial speed is the square root of the intercept.
- C2. Plot $E_{\max}$ against f ; gradient = h ; intercept on the energy axis = $-\phi$, so the work function is minus the intercept. (The intercept on the *frequency* axis is the threshold frequency $f_0 = \phi/h$.)
- C3. Dividing by t gives $\frac{s}{t} = u + \frac{1}{2}at$. Plot s/t against t ; gradient = $\frac{1}{2}a$; intercept = u . Plotting s against t would give a curve, and plotting s against t^2 only works when $u = 0$.
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D *getting the constant out*

- D1. Gradient = $\frac{4\pi^2}{g}$, so

$$g = \frac{4\pi^2}{\text{gradient}} = \frac{4\pi^2}{4.05} = 9.75 \text{ m s}^{-2}.$$

- D2. Plot L against $1/f$. Gradient $= \frac{v}{4}$, so the speed of sound is four times the gradient. The intercept on the L axis is $-e$, so the end correction is minus the intercept: a quantity you could not measure directly at all, recovered from where the line crosses.
- D3. Rearranging, $\frac{1}{v} = -\frac{1}{u} + \frac{1}{f}$. Plot $1/v$ against $1/u$. The gradient is -1 , a pure number fixed by the algebra, so it tells you nothing about the lens and serves only as a check that the data behave. All the physics is in the intercept, which is $1/f$, giving $f = 1/\text{intercept}$.